OPERATING SYSTEMS (CS304)

Disk Scheduling Algorithms

Learning Objectives

Understand disk scheduling, disk access optimization, and major scheduling algorithms such as FCFS, SSTF, SCAN, C-SCAN, LOOK, and C-LOOK.

Introduction

Disk scheduling algorithms determine the order in which disk I/O requests are serviced. Their primary goal is to reduce seek time and improve overall disk performance.

Need for Disk Scheduling

• Minimize seek time
• Reduce disk access latency
• Improve throughput
• Enhance system performance
• Efficient utilization of storage devices

Key Terms

Seek Time: Time taken to move the disk arm to the required track.
Rotational Latency: Time waiting for the desired sector to rotate under the read/write head.
Transfer Time: Time required to transfer data.

FCFS Disk Scheduling

First Come First Serve (FCFS) services requests in the order they arrive.

Advantages of FCFS

Simple, fair, and easy to implement.

Disadvantages of FCFS

May result in high seek time and poor performance.

SSTF (Shortest Seek Time First)

SSTF selects the request closest to the current head position, reducing seek time.

Advantages of SSTF

Better performance than FCFS and lower average seek time.

Disadvantages of SSTF

May cause starvation for requests far from the current head position.

SCAN Algorithm

SCAN moves the disk arm in one direction servicing requests until the end is reached, then reverses direction.

C-SCAN Algorithm

Circular SCAN services requests in one direction only and returns to the beginning without servicing requests during the return trip.

LOOK Algorithm

LOOK is similar to SCAN but only travels as far as the last request in a direction before reversing.

C-LOOK Algorithm

C-LOOK is similar to C-SCAN but jumps directly to the first pending request instead of moving to the physical end of the disk.

Numerical Example

Request Queue: 98, 183, 37, 122, 14, 124, 65, 67

Initial Head Position: 53

Different algorithms produce different total seek distances and performance results.

Comparison of Disk Scheduling Algorithms

FCFS: Simple but inefficient.
SSTF: Better seek time but starvation possible.
SCAN: Good overall performance.
C-SCAN: Uniform waiting time.
LOOK: Reduced unnecessary movement.
C-LOOK: Efficient and balanced performance.

Applications

Disk scheduling algorithms are used in operating systems, database servers, storage controllers, and enterprise data centers.

Exam Important Questions

1. Define Disk Scheduling.
2. Explain FCFS Disk Scheduling.
3. Explain SSTF Algorithm.
4. Differentiate SCAN and C-SCAN.
5. Explain LOOK and C-LOOK.
6. Solve numerical problems on Disk Scheduling.

Summary

Disk scheduling algorithms optimize storage access by reducing seek time and improving performance. FCFS, SSTF, SCAN, C-SCAN, LOOK, and C-LOOK each offer different trade-offs in efficiency and fairness.